Manvir Dhami

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Education

University of British Columbia — Vancouver, BC
B.A.Sc. in Computer Engineering, 3.8 GPA, EIT-eligible (EGBC) — *Sep 2020 – May 2024*

Technical Skills

Electrical and Controls: schematic and PCB design (KiCad), bills of materials, low-voltage systems, opto-isolated I/O, power switching (MOSFET), current and temperature sensing, wire-to-board connectors, bench test equipment (multimeter, oscilloscope, power supplies), PLC ladder logic (IEC 61131-3, OpenPLC)

Languages and Embedded: Python, C, Rust, Verilog/VHDL, assembly, Bash; ESP32/ARM microcontrollers, FPGA

Cloud and DevOps: AWS (ECS, EC2, S3), Docker, Linux, Git, CI/CD (GitHub Actions), Jira/Agile

Projects

Low-Voltage Charge-Bay Controller | *KiCad, ESP32, Python, IEC 61131-3 (in progress, schematic complete)*

- Designing a four-channel LiPo charging bay with per-channel current sensing (INA226) and pack temperature monitoring (NTC), where the over-temperature cutoff is a hardware comparator rather than firmware so it still trips even if the software hangs.
- Designing high-side P-MOSFET switching with flyback protection and fail-safe gate pull-ups. The design also includes opto-isolated inputs, an E-stop chain, wire-to-board connectors and silkscreened test points.
- Writing the control sequencing in OpenPLC ladder logic, with a Python plant simulator for fault injection and a nine-test validation protocol that records measured pass/fail values.
- Documenting the design with a bill of materials and a connector pinout and wiring table.

Turing-Complete Processor | *Verilog*

- Designed a Turing-complete processor in Verilog, implementing the datapath, control unit and instruction set, and verified every instruction (including branching) using a testbench-driven test suite.
- Wrote an assembler for the instruction set and brought the design up on an FPGA to run small programs.

Experience

Software Engineer / Project Lead — May 2024 – Present
Korotu Technology — *Vancouver, BC*

- *Lead development of the company's drone survey and photogrammetry platform for 30+ conservation and forestry clients, handling software, project delivery and cloud infrastructure.*
- **Test, Validation and Specification**
  - Wrote 21 platform specifications with feature-level acceptance criteria and maintained the defect log to track issues before release.
  - Rewrote the platform's allometric sampling method and cut tree-diameter estimation error by 27% (26.4 to 19.3 cm), lifting crown-detection accuracy from 0.32 to 0.55 F1 in measured tests.
  - Built GitHub Actions tests that check platform accuracy and catch regressions before each release.
- **Systems and Operations**
  - Operated field hardware (drones, batteries, sensors, cabling) across client deployments and flight operations.
  - Investigated a failed drone survey across 16 configurations, identified image quality as the limiting factor rather than the software, and implemented a fallback.
  - Moved 10 TB of data and 25 servers from Azure to AWS and switched legacy hosting to on-demand ECS with automated CI/CD, cutting annual infrastructure costs by 70% (from $20,000 to $6,000).
  - Migrated databases across clouds with zero downtime for 50+ users.
- **Leadership and Documentation**
  - Mentored five co-op engineers through weekly planning and code review.
  - Led technical delivery across six spatial data platforms, coordinating design, implementation and deployment across multiple teams.
  - Prepared test plans and accuracy reports for each project, documenting test results and any gaps against requirements.
  - Wrote the company's documentation standard and developed infrastructure and security onboarding materials for new hires.